11 Chemistry Unit 2 Topic Test 2: Volumetric Analysis and Limiting Reactants — Worked Solutions and Marking Guide

TEACHER MARKING GUIDE — not for distribution to students

Total 40 marks: Section A 10 × 1 mark (separate multiple-choice answer sheet), Section B 30 marks. Writing time 46 minutes (40 marks × 1.15 min per mark). Students use the VCE Chemistry Data Book (SLC 25 °C and 100 kPa; V_m = 24.8 L mol^{-1} at SLC; R = 8.31 J mol^{-1} K^{-1}; 1 atm = 101.3 kPa). **One mark per idea or step; no half marks.** Units and significant figures are marked once, at the final answer of each part. **Rounding tolerance:** accept a final answer that differs only in the last significant figure because intermediate values were rounded; wider bands are stated where needed. **Consequential marking:** where a part uses an earlier answer, award full marks for correct working from the student's own earlier value (shown as *Consequential*). Each item is labelled with its difficulty (Easy / Medium / Hard, by marks: 10 / 20 / 10) and demand level (L1 recall, L2 routine procedure, L3 interpret or explain, L4 evaluate or justify).

Section A — Multiple-choice answer key

Key balance: A × 3, B × 2, C × 3, D × 2 (key string C A D B A C B D C A). Every distractor is the answer produced by one named slip, so tallies of wrong letters show which misconceptions to reteach.

Q	Key	Working for the key	Distractor diagnosis
1	**C**	**Easy (L1).** Titre = final burette reading − initial burette reading: the volume delivered from the burette.	A: confuses the titre with the aliquot; B: took the final reading as the titre (did not subtract the initial reading); D: confuses a titre with the mean concordant titre.
2	**A**	**Easy (L2).** n = c × V = 0.0800 × 0.01750 = 1.40 × 10^{-3} mol	B: volume left in mL; C: divided V by c (0.01750 ÷ 0.0800); D: divided c by V (0.0800 ÷ 0.01750).
3	**D**	**Easy (L1).** Read the bottom of the meniscus at eye level to avoid parallax error; burette readings are recorded to two decimal places.	A: reads the wrong part of the meniscus; B: parallax: looking from above gives a wrong reading; C: does not know burettes are read to ±0.01–0.05 mL (two decimal places).
4	**B**	**Medium (L2).** n(NaOH) = 0.112 × 0.01875 = 2.10 × 10^{-3} mol; n(acid) = ½ × 2.10 × 10^{-3} = 1.05 × 10^{-3} mol; c = 1.05 × 10^{-3} ÷ 0.02500 = 0.0420 M	A: ignored the 1 : 2 mole ratio; C: multiplied by 2 instead of dividing (ratio inverted); D: swapped the aliquot and titre volumes.
5	**A**	**Medium (L2).** 0.10 mol P_4 needs 0.60 mol Cl_2; only 0.50 mol present, so Cl_2 is limiting. n(PCl_3) = 0.50 × 4 ÷ 6 = 0.33 mol	B: assumed P_4 is limiting because it has the smaller amount (did not use the ratio); C: used a 1 : 1 ratio between Cl_2 and PCl_3; D: inverted the Cl_2 : PCl_3 ratio (0.50 × 6 ÷ 4).
6	**C**	**Medium (L2).** n($TiCl_4$) = 760 000 ÷ 189.9 = 4.00 × 10^3 mol; n(Mg) = 180 000 ÷ 24.3 = 7.41 × 10^3 mol. $TiCl_4$ would need 8.00 × 10^3 mol Mg, so Mg is limiting. n(Ti) = ½ × 7.41 × 10^3 = 3.70 × 10^3 mol; m = 3.70 × 10^3 × 47.9 = 1.77 × 10^5 g = 177 kg	A: compared amounts without the 1 : 2 ratio, so chose $TiCl_4$; B: correct limiting reactant but used a 1 : 1 ratio for Mg : Ti; D: applied the 2 : 1 ratio twice (divided by 4).
7	**B**	**Hard (L3).** n($TiCl_4$) reacted = ½ × 7407 = 3704 mol; remaining = 4002 − 3704 = 298 mol; m = 298 × 189.9 = 5.67 × 10^4 g = 56.7 kg (accept 56–57 kg from rounding)	A: gave the mass of $TiCl_4$ that reacted, not the mass remaining; C: subtracted the mass of Mg from the mass of $TiCl_4$; D: gave the amount remaining (298 mol) without multiplying by M.
8	**D**	**Easy (L1).** The rough titre shows roughly where the end point is, so accurate titrations can be added dropwise near it.	A: thinks the rough titre is averaged with the accurate titres; B: confuses the rough titration with checking glassware; C: confuses the rough titration with standardisation.
9	**C**	**Hard (L4).** Acid lost from the flask needs less NaOH: smaller titre, smaller calculated n(HCl), so the calculated concentration is too low. The other three options (overshooting, a pipette that over-delivers, an undelivered hanging drop) all make the calculated concentration too high.	A: direction reversed: overshooting gives a larger titre, so a higher result; B: more acid in the flask gives a larger titre, so a higher result; D: the recorded titre includes NaOH that was not delivered, so a higher result.

10	**A**	**Medium (L3).** n(HNO_3) = 0.200 × 0.0250 = 5.00 × 10^{-3} mol; n(KOH) = 0.300 × 0.0200 = 6.00 × 10^{-3} mol; 1 : 1, so KOH is in excess by 1.00 × 10^{-3} mol	B: judged by the larger volume, not the amount in mol; C: gave the total amount of KOH, not the excess; D: subtracted the concentrations (0.300 − 0.200).

Section B — Short-answer worked solutions

Each part shows the worked solution, then the mark allocation (one mark per step or idea), consequential notes and answers to reject.

Question 1 — Checking an acid batch (9 marks)

a) (1 mark)

30.25 − 1.20 = 29.05 mL

1 mark 29.05 mL. *[Easy; L2]*

Reject: 30.25 mL (final reading only); 31.45 mL (readings added).

b) (2 marks)

The titres are 29.30 (rough), 28.70, 29.05 and 28.75 mL. Only 28.70 mL and 28.75 mL agree within 0.10 mL; 29.05 mL is not concordant and the rough titre is not used. There are not yet three concordant titres.

Grace should carry out more titrations until she has three concordant titres.

1 mark identifies that there are not three concordant titres (29.05 mL does not agree with 28.70 and 28.75 mL). **1 mark** do further titrations until three concordant titres are obtained. *[Medium; L3]*

Consequential: Allow consequential use of the student's titre for titration 2 from part a).

Reject: 'Average 28.70 and 28.75' (only two concordant titres; the stated rule needs three); 'average all four titres'; 'repeat the rough titration'.

c) (4 marks)

n(NaOH) = 0.165 × 0.02875 = 4.74 × 10^{-3} mol

n(HCl) = n(NaOH) = 4.74 × 10^{-3} mol (1 : 1)

c(HCl) = 4.74 × 10^{-3} ÷ 0.02500 = 0.190 M

0.190 M is above 0.185 M, so the batch does not meet the requirement (it is too concentrated).

1 mark n(NaOH) = 4.74 × 10^{-3} mol. **1 mark** 1 : 1 ratio, n(HCl) = 4.74 × 10^{-3} mol. **1 mark** c = 0.190 M (accept 0.189–0.190 M). **1 mark** judgement consistent with the calculated value and the 0.175–0.185 M range. *[Medium; L3]*

Consequential: Each step is consequential on the previous one; the judgement mark is consequential on the student's concentration.

Reject: 0.143 M (aliquot and titre volumes swapped); 4.74 × 10^{-3} M (amount given as concentration); a judgement with no reference to the range.

d) (2 marks)

The titre would be only about 28.75 × 10.00 ÷ 25.00 = 11.5 mL.

Each burette reading has the same uncertainty (about ±0.02–0.05 mL), which is a larger percentage of a smaller titre, so the result would be less precise. (Accept: the pipette's uncertainty is also a larger percentage of a smaller aliquot.)

1 mark the titre would be smaller (about 11.5 mL). **1 mark** so the fixed reading uncertainty of the burette (or pipette) is a larger percentage of the measurement: a less precise result. *[Hard; L4]*

Reject: 'The answer would be wrong / the concentration would change' (the calculated concentration is the same in principle); 'it would take longer'; 'smaller volumes are less accurate' with no reason.

Part a) reads the burette table; part b) deliberately leaves only two concordant titres; part c) supplies the mean, so the calculation can be marked independently. Part d) is about measurement uncertainty, not a technique error.

Question 2 — Making urea fertiliser (9 marks)

a) (1 mark)

n = 340 000 g ÷ 17.0 g mol^{-1} = 2.00 × 10^4 mol

1 mark 2.00 × 10^4 mol. *[Easy; L2]*

Reject: 20.0 mol (kg not converted to g).

b) (2 marks)

n(CO_2) = 520 000 ÷ 44.0 = 1.18 × 10^4 mol

To react all the CO_2 would need 2 × 1.18 × 10^4 = 2.36 × 10^4 mol NH_3, but only 2.00 × 10^4 mol is present, so NH_3 is limiting.

(Or: 2.00 × 10^4 mol NH_3 needs only 1.00 × 10^4 mol CO_2, and 1.18 × 10^4 mol is present.)

1 mark n(CO_2) = 1.18 × 10^4 mol. **1 mark** comparison that uses the 2 : 1 ratio and concludes NH_3 is limiting. *[Medium; L2]*

Consequential: On part a).

Reject: 'NH_3 is limiting because its mass is smaller'; a comparison of amounts that ignores the 2 : 1 ratio.

c) (2 marks)

n(urea) = ½ × n(NH_3) = ½ × 2.00 × 10^4 = 1.00 × 10^4 mol

M($CO(NH_2)_2$) = 12.0 + 16.0 + 2(14.0 + 2.0) = 60.0 g mol^{-1}; m = 1.00 × 10^4 × 60.0 = 6.00 × 10^5 g = 600 kg

1 mark n(urea) = 1.00 × 10^4 mol (from the limiting reactant, 1 : 2). **1 mark** 600 kg. *[Medium; L2]*

Consequential: On the student's limiting reactant.

Reject: 1200 kg (ratio ignored); 709 kg (calculated from CO_2).

d) (2 marks)

n(CO_2) reacted = 1.00 × 10^4 mol; remaining = 1.182 × 10^4 − 1.00 × 10^4 = 1.82 × 10^3 mol

m = 1.82 × 10^3 × 44.0 = 8.00 × 10^4 g = 80.0 kg

1 mark amount of CO_2 remaining, 1.82 × 10^3 mol. **1 mark** 80.0 kg (accept 79.9–80.2 kg). *[Hard; L3]*

Consequential: On parts b) and c).

Reject: 440 kg (mass of CO_2 reacted); 180 kg (subtracted masses 520 − 340).

e) (2 marks)

All of the ammonia is used up (it is limiting), so extra ammonia can react with the carbon dioxide left over and form more urea.

Carbon dioxide is already in excess, so adding more of it only leaves more unreacted CO_2: the amount of urea is still set by the ammonia.

1 mark NH_3 is limiting, so extra NH_3 reacts with leftover CO_2 to make more urea. **1 mark** CO_2 is already in excess, so extra CO_2 cannot react (urea is limited by NH_3). *[Medium; L3]*

Reject: 'Ammonia has a smaller molar mass'; 'more reactant always makes more product'.

Part e) is reverse reasoning: students justify a given statement from their limiting-reactant result.

Question 3 — Precise or accurate? (6 marks)

a) (1 mark)

Callum (his titres agree within 0.05 mL; Harriet's spread over 0.50 mL).

1 mark Callum. *[Easy; L1]*

Reject: Harriet (her mean equals the teacher's value: that is accuracy, not precision).

b) (2 marks)

When the bubble is pushed out, the liquid level in the burette falls by the volume of the bubble as well as by the volume of NaOH delivered.

So each recorded titre (final − initial reading) includes the bubble's volume and is larger than the volume of NaOH that actually entered the flask.

1 mark the burette reading changes by the bubble's volume as well as the NaOH delivered. **1 mark** so the recorded titre is larger than the volume of NaOH that actually reacted. *[Medium; L3]*

Reject: 'The bubble added extra NaOH'; 'the bubble diluted the NaOH'; 'air reacts with the acid'.

c) (1 mark)

Systematic error.

1 mark systematic. *[Easy; L1]*

Reject: random; 'human error'; 'mistake' alone.

d) (2 marks)

It would not improve the accuracy.

His air-bubble error is systematic: every titre is too high by about the same amount, so the mean of eight titres is still too high. Averaging reduces only the effect of random errors. He must remove the bubble from the burette tip before taking the initial reading.

1 mark would not improve the accuracy (committed judgement). **1 mark** reason: a systematic error shifts every titre in the same direction, and averaging only reduces the effect of random error (or: he must fix the method, e.g. run some solution through the tap to clear the bubble before reading the burette). *[Hard; L4]*

Reject: 'Yes, more results make it more accurate'; 'averaging removes random error' (averaging reduces the effect of random error, it does not remove it).

Precision (agreement between results) vs accuracy (closeness to the true value). The teacher's mean is the reference value.

Question 4 — How much vitamin C? (6 marks)

a) (1 mark)

$n(NaOH) = 0.105 \times 0.02660 = 2.79 \times 10^{-3}$ mol

1 mark 2.79×10^{-3} mol. *[Easy; L2]*

Reject: 2.79 mol (volume in mL).

b) (2 marks)

$n(C_6H_8O_6) = n(NaOH) = 2.79 \times 10^{-3}$ mol (1 : 1)

$M(C_6H_8O_6) = 6(12.0) + 8(1.0) + 6(16.0) = 176.0$ g mol^{-1}; $m = 2.793 \times 10^{-3} \times 176.0 = 0.492$ g = 492 mg

1 mark 1 : 1 ratio, $n(C_6H_8O_6) = 2.79 \times 10^{-3}$ mol. **1 mark** 492 mg (accept 491–492 mg). *[Medium; L2]*

Consequential: On part a).

Reject: 0.492 mg or 4.92×10^5 mg (unit conversion); 500 mg (the label value).

c) (1 mark)

492 ÷ 500 × 100 = 98.3%

1 mark 98.3% (accept 98.2–98.4%). *[Easy; L2]*

Consequential: On part b).

Reject: 102% (inverted).

d) (2 marks)

The teacher is correct. Adding more water does not change the amount (mol) of ascorbic acid in the flask.

The volume of NaOH needed depends on the amount (mol) of ascorbic acid to be neutralised, not on its concentration, so the titre is unchanged.

1 mark the teacher is correct because the amount (mol) of ascorbic acid is unchanged by the extra water. **1 mark** the titre depends on the amount of acid (1 : 1 mole ratio), not its concentration, so it is the same. *[Hard; L4]*

Reject: 'Water does not react' alone (no link to amount); 'the concentration is the same' (it is not: it is lower); 'the titre is larger because the solution is more dilute'.

Part d) is reverse reasoning: students justify a statement about dilution in the conical flask.

Mark summary

Section	Marks	Notes
A — multiple choice	10	10 × 1 mark; key A3 / B2 / C3 / D2
B — short answer	30	Q1 9, Q2 9, Q3 6, Q4 6
Total	**40**	Writing time 46 minutes

Difficulty tally (by marks)

Section	Easy	Medium	Hard	Total
A — multiple choice	4	4	2	10
B — short answer	6	16	8	30
Total	**10**	**20**	**10**	**40**
Target (25 / 50 / 25%)	10	20	10	40

Difficulty and demand by item

Item	Marks	Difficulty	Demand
A Q1	1	Easy	L1
A Q2	1	Easy	L2
A Q3	1	Easy	L1
A Q4	1	Medium	L2
A Q5	1	Medium	L2
A Q6	1	Medium	L2
A Q7	1	Hard	L3
A Q8	1	Easy	L1
A Q9	1	Hard	L4
A Q10	1	Medium	L3
B Q1a)	1	Easy	L2
B Q1b)	2	Medium	L3
B Q1c)	4	Medium	L3
B Q1d)	2	Hard	L4
B Q2a)	1	Easy	L2
B Q2b)	2	Medium	L2
B Q2c)	2	Medium	L2
B Q2d)	2	Hard	L3
B Q2e)	2	Medium	L3
B Q3a)	1	Easy	L1
B Q3b)	2	Medium	L3
B Q3c)	1	Easy	L1
B Q3d)	2	Hard	L4
B Q4a)	1	Easy	L2
B Q4b)	2	Medium	L2
B Q4c)	1	Easy	L2
B Q4d)	2	Hard	L4

Cognitive demand

Demand level	Marks	Share
L1 recall	5	12%
L2 routine procedure	14	35%
L3 interpret / explain	14	35%
L4 evaluate / justify	7	18%

Routine marks (L1 + L2): 19 of 40 (48%).

Key knowledge assessed

• acid–base titrations (direct only): titres, aliquots, concordant titres, stoichiometry

• titration technique (burette reading, rough and concordant titres) and error analysis

• limiting and excess reactants: identify, mass of product, excess remaining

Not assessed (not taught this year): indicators and indicator choice, pH calculations, back titrations, calibration curves, solubility curves, colorimetry, gravimetric analysis.